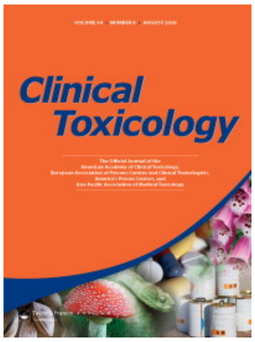

Clinical Toxicology

ISSN: 1556-3650 (Print) 1556-9519 (Online) Journal homepage: www.tandfonline.com/journals/ictx20

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To cite this article: Chen Chen, Yuhao Yuan, Wenshuai Tan, Xiangting Gao, Zhonghao Yu, Jiaqi Liu, Yiwu Zhou & Xiaohong Zhao (10 Aug 2026): Acute thallium poisoning: clinical features, toxicokinetics and one-year neurological outcomes, Clinical Toxicology, DOI: 10.1080/15563650.2026.2697048

To link to this article: https://doi.org/10.1080/15563650.2026.2697048

CLINICAL TOXICOLOGY
https://doi.org/10.1080/15563650.2026.2697048

RAPID COMMUNICATION

Acute thallium poisoning: clinical features, toxicokinetics and one-year neurological outcomes

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ABSTRACT

Introduction: Thallium poisoning is a rare but highly toxic heavy metal intoxication associated with peripheral neuropathy. However, longitudinal data integrating toxicokinetic profiles with electrophysiological outcomes remain limited.

Methods: Five patients with confirmed acute thallium poisoning were evaluated. Clinical symptoms, laboratory results, dynamic serum and urinary thallium concentrations, and electrophysiological results were recorded. All patients underwent electrophysiological follow-up one year later. Thallium elimination kinetics were analyzed using nonlinear regression.

Results: All patients developed peripheral neuropathy. Sensory symptoms generally preceded the onset of alopecia. During treatment, dynamic monitoring showed a progressive decrease in thallium concentrations. Toxicokinetic analysis suggested an approximate first-order elimination pattern, with an estimated serum half-life ranging from 3.8 to 5.7 days. Electrophysiological studies revealed distal axonal neuropathy characterized by reduced compound muscle action potential amplitudes. At one-year follow-up, residual axonal involvement persisted in some patients.

Discussion: Peripheral neuropathic symptoms may serve as an early clinical indicator of thallium poisoning. Persistent electrophysiological abnormalities despite declining thallium levels suggest incomplete neurological recovery.

Conclusions: Dynamic monitoring of thallium levels can help evaluate elimination kinetics during detoxification therapy. Persistent axonal injury suggests that thallium-induced neuropathy may result in long-term neurological sequelae, highlighting the importance of early diagnosis and long-term electrophysiological monitoring.

ARTICLE HISTORY
Received 5 April 2026
Revised 13 June 2026
Accepted 22 June 2026

KEYWORDS
Thallium poisoning; toxicokinetics; peripheral neuropathy; electromyography; follow-up study

Introduction

Previous studies have reported the typical triad of acute thallium poisoning, including gastrointestinal symptoms, peripheral neuropathy, and delayed alopecia [1,2]. However, early manifestations are often atypical, leading to delayed diagnosis and treatment [3]. Standard treatment strategies include hemoperfusion, Prussian blue, and chelation therapy [4], which may accelerate thallium clearance [5,6]. Nevertheless, long-term neurological outcomes remain unclear, and longitudinal data linking toxicokinetics with electrophysiological recovery are limited. Therefore, we describe a cluster of acute thallium poisoning cases, focusing on clinical features, elimination kinetics, and one-year electrophysiological outcomes.

Methods

Five patients with confirmed acute thallium poisoning from a common-source exposure were included. Patients resided in the same geographic area but belonged to different households. Epidemiological investigation suggested that exposure occurred through ingestion of chili paste contaminated with thallium, although the exact timing and quantity of ingestion could not be reliably determined. Due to the nonspecific nature of early symptoms, the diagnosis was delayed. The index patient was diagnosed approximately 15 days after symptom onset, following toxicological testing prompted by persistent and unexplained neurological symptoms. Subsequent identification of additional cases from the same exposure source was achieved through clinical

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/15563650.2026.2697048.

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and epidemiological investigation. Diagnosis was based on consistent clinical presentation and elevated thallium concentrations in serum and urine samples measured by inductively coupled plasma mass spectrometry (ICP-MS). Nerve conduction studies and needle electromyography (EMG) were performed using a standard EMG system in routine clinical practice.

Thallium elimination kinetics were analyzed using a first-order exponential decay model fitted to serial serum and urinary thallium measurements. Parameters, including the elimination rate constant and half-life, were estimated using nonlinear least-squares regression.

Results

Clinical characteristics

The basic information and clinical characteristics of the five patients are summarized in Table 1. Alopecia and limb pain or numbness were the most common manifestations. None of the patients reported gastrointestinal symptoms. Physical examination suggested that potential peripheral nerve injury was likely present in all patients. The median duration from symptom onset to clinical presentation was 19.5 days (range 2–50 days) for alopecia and 15.5 days (range 15–27 days) for limb pain or numbness. In most patients, sensory symptoms had been present for a longer period than alopecia before consultation.

Laboratory findings

Serum and urinary thallium concentrations were markedly elevated in all patients, greatly exceeding the normal reference ranges. Detailed thallium concentration data are provided in the Supplementary material.

Toxicokinetics

All patients underwent hemoperfusion therapy to enhance thallium elimination, together with oral Prussian blue and dimercaptosuccinic acid (DMSA) after the diagnosis. Serial monitoring of serum and urinary thallium concentrations was performed in three patients during treatment (Figure 1). Following the initiation of therapy, thallium concentrations showed a continuous decreasing trend. Serum thallium levels decreased to below 100 μg/L (489 nmol/L) within 6–10 days. Exponential decay curves demonstrated good fit for serum thallium kinetics, yielding elimination rate constants (*k*) between 0.12 and 0.18 day^{-1} and corresponding serum elimination half-lives of 3.8–5.7 days. The total exposure burden, quantified by the area under the concentration–time curve (AUC), was highest in patient 2 (2,826 μg day/L). Urinary thallium levels dropped below 100 μg/g creatinine (approximately 55 μmol/mol creatinine) between days 14 and 15. Urinary kinetics exhibited greater variability at lower concentrations. Therefore, the elimination pattern of urinary thallium was described using descriptive parameters rather than kinetic modeling.

Neurology examination findings

Electromyography revealed reduced compound muscle action potential (CMAP) amplitude (AMP) in distal muscles in all patients. Meanwhile, nerve conduction velocity (NCV) remained generally normal. The above electrophysiological findings were consistent with distal axonal neuropathy. The EMG results are summarized in the Supplementary material.

Follow-up results

At one-year follow-up, most patients showed improvement in clinical symptoms. However, three patients continued to exhibit reduced CMAP amplitude (< 5 mV) at the distal stimulation sites, indicating persistent axonal involvement and incomplete neurological recovery. Additional details are provided in the Supplementary material.

Table 1. Demographics, clinical manifestations, and duration of symptoms from onset to presentation.

Patient no.	Sex	Age (years)	Alopecia (days)	Limb pain/numbness (days)	Hyperalgesia of the extremities	Other symptoms
1	F	66	12	15	Y	None
2	F	65	None	15	Y	Blurred vision in both eyes
3	M	66	2	16	N	Diminished Achilles tendon reflex
4	M	66	27	27	Y	Tenderness on palpation of the gastrocnemius muscles in both lower limbs
5	F	57	50	None	Y	None

F: female; M: male; Y: yes (the symptom is present); N: no (the symptom is not present).

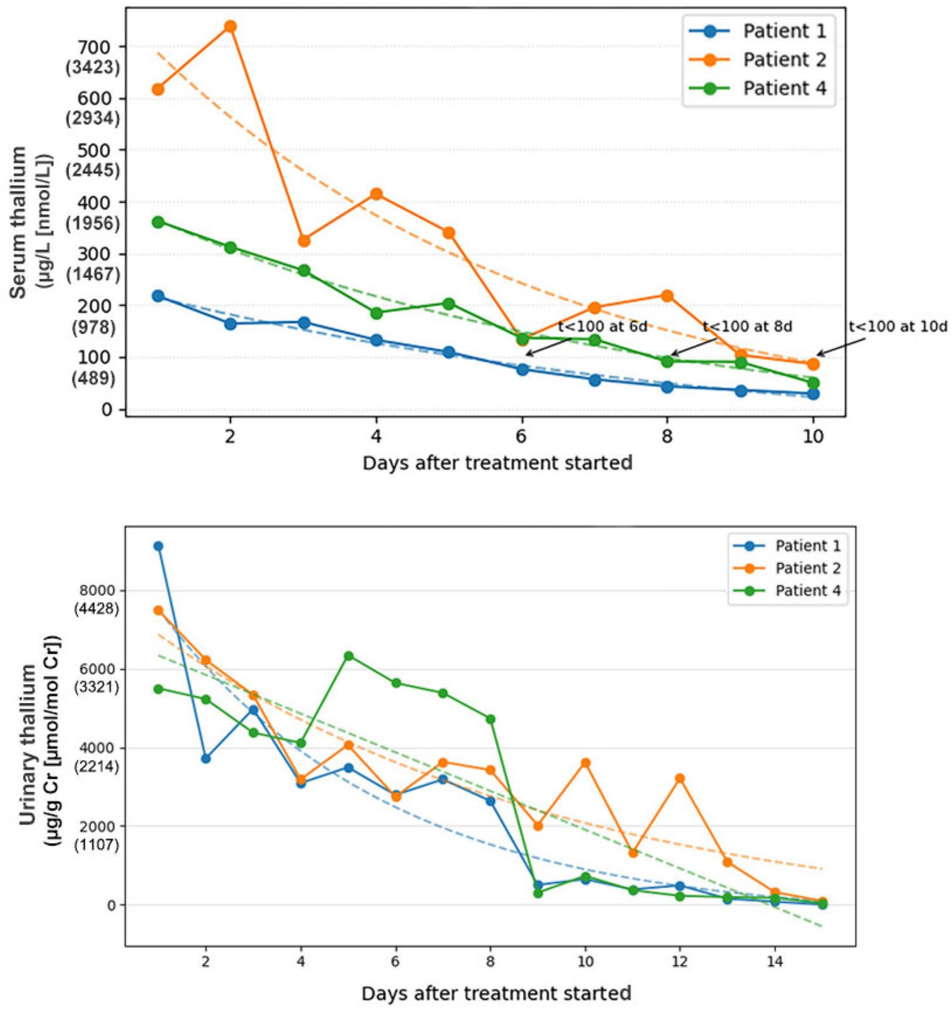

Figure 1. Dynamic changes in serum and urinary thallium concentrations during treatment. Serum and urinary thallium concentrations in patients 1, 2, and 4 were plotted across the treatment period. Day 0 represents the initiation of combined treatment, including hemoperfusion, Prussian blue and succimer administered according to standard clinical practice. Solid lines represent actual measured values, while dashed lines indicate nonlinear regression-fitted exponential decay curves used to estimate elimination kinetics (curve stability decreased at lower concentrations). The notation $t < 100$ indicates that serum thallium concentrations decreased below 100 μg/L. The threshold of 100 μg/L is displayed to indicate reduction toward safer levels. AUC values and estimated half-lives were derived from these fitted curves to reflect total toxic burden and treatment effectiveness.

Discussion

Sensory disturbances generally preceded alopecia in our patients, suggesting that peripheral neuropathy symptoms may be an important early clue to thallium poisoning. None of the patients reported gastrointestinal symptoms, which may be related to delayed presentation and recall bias. Electrophysiological evaluation demonstrated reduced CMAP amplitudes in distal muscles, indicating peripheral neuropathy.

Serum and urinary thallium concentrations declined progressively during treatment. Serum thallium elimination follows a predictable first-order kinetic pattern under combined hemoperfusion and chelation therapy, and nonlinear regression of the elimination curves yielded estimated serum thallium half-live of 3.8–5.7 days. The observed elimination kinetics likely reflect treatment-enhanced clearance rather than intrinsic thallium toxicokinetics alone. The estimated half-life is consistent with previous reports, which show that combined detoxification therapies can substantially shorten the apparent elimination half-life of thallium compared with passive elimination [6,7]. Higher cumulative exposure (larger AUC) indicates a greater thallium burden *in vivo*, and is associated with slower elimination and delayed achievement of clinically safe Tl levels. Urinary elimination showed greater variability at low concentrations, possibly because of renal handling, urine dilution, and increased analytical uncertainty near the lower limits of quantification [8,9].

At one-year follow-up, residual axonal involvement persisted in some patients. The finding highlights the limited regenerative capacity of axons after heavy-metal toxicity, suggesting that thallium-induced neuropathy may lead to long-term, partially irreversible axonal injury.

Conclusion

This study demonstrates that peripheral neuropathical symptoms may precede alopecia and serve as an important early indication of thallium poisoning. Dynamic monitoring of serum and urinary thallium concentrations enables visualization of treatment-associated elimination kinetics and helps assess treatment efficacy. Persistent axonal involvement at one-year follow-up indicates that neurological recovery may be incomplete, suggesting that thallium-induced peripheral nerve damage may have long-term effects. Delayed recognition of thallium poisoning may contribute to persistent neurological sequelae, highlighting the importance of clinical awareness and long-term neurological follow-up.

Consent form

Written informed consent was obtained from all patients.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This study did not receive any specific grant from any funding agency.

Data availability statement

Data are available from the corresponding author upon reasonable request. Due to the inclusion of potentially identifiable clinical information, the data are not publicly available.

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